

LEGISLATIVE CRAWLER

Technical Bridge Document: Systemic Architecture & Narrative Implementation

1. EXECUTIVE ENGINE SUMMARY

This technical blueprints document serves as a systemic bridge guide for game architects, systems designers, and engineers. It details how to translate ideological positioning scales directly into core structural data loops, shifting a Pokémon-style retro engine into an active, grid-based constitutional simulation environment. The core gameplay loops center entirely around structured, technical mechanics mapping the legislative process onto classic tactical RPG constructs.

1.1 The Polarization Scale Engine

The game tracking system treats ideological spaces as a bounded numerical timeline array $[0, 100]$, functioning as a global balancing modifier. Left-leaning factions map to integers $0 \leq x \leq 49$, Right-leaning factions map to integers $51 \leq x \leq 100$, and absolute bipartisanship rests at center equilibrium $x = 50$. When a user selects a political party, their baseline configuration establishes them within a structurally persistent parliamentary minority. Progression relies on targeted data interactions with localized individual actors or collective node networks.

Architectural Rule: Mathematical Polarization Metric

Let B_v be the Base Bill Value and N_r be the Target Representative Value. The delta calculation $D = |B_v - N_r|$ dictates the core programmatic difficulty layer, mapping options to user selection loops and modifying tile mechanics.

2. TECHNICAL SYSTEM TRANSITIONS

2.1 Input & Logic Array: The Intra-Party Mechanics

Interactions with members of the user's primary party utilize dynamic multiple-choice arrays based on the verified 100-Question U.S. Citizenship Naturalization Exam. Complexity functions dynamically relative to the delta matrix:

- **Low Polarization Array (Moderate Nodes, Close to 50):** The UI populates a strict, clear 3-choice multiple-choice grid.
- **High Polarization Array (Extremist Nodes, Near 5 or 95):** The system renders an exhaustive, confusing 10-to-12 choice grid matrix. Wrong answers populate with highly similar semantic phrasing to penalize guess patterns.

2.2 Input & Logic Array: Cross-Aisle Interrogations

Engaging members of the opposition party switches the gameplay engine from a multiple-choice array to an open-ended keyboard text entry mode.

- **Moderate Opposition (Rating ~55/45):** Displays clear textual presentation accompanied by programmatic dialogue hints providing localized assistance.

- **Extreme Opposition (Rating ~95/5):** Implements visual text degradation effects. Text boxes render scrambled, obscured, or muddled with procedural text strings (e.g., " . . . "), omitting crucial contextual keywords.

The text box parsing engine executes fuzzy text matching configurations, matching strings via case-insensitive logic while checking distance criteria to allow minor Chromebook keyboard slip errors without breaking execution chains.

2.3 Faction Structure & Strategic Compounding

To preserve technical pacing, the 435 individual House nodes group into persistent structural entities called **Caucuses / Coalitions**. Interacting directly with a designated Coalition Leader node executes a macro-transaction: winning the trivia encounter locks down the voting power block for that entire faction cluster, dynamically adding to the global player's `Vote_Counter` variable.

Faction Entity Type	Ideological Range	Vote Block Weight	Combat Architecture Mapping
Bipartisan Solvers Caucus	48 – 52	40 Votes	3-Choice Grid / Open-ended with clean directional hints
Party Loyalist Factions	30s / 70s	25 Votes each	Mid-tier 5-to-6 Choice Grid UI configuration
Ideological Wings	5–10 / 90–95	15 Votes each	12-Choice Exhaustive Grid / Scrambled, non-hint Text Entry

2.4 The Automated Procedural Gate (The 175-Vote Trigger)

When the global counter satisfies the threshold value `Vote_Counter = 175` (exactly 20% clear of the required **218** majoritarian threshold), the system triggers an un-skippable boss ambush event by an Extremist Faction Leader. This event is structurally hardcoded as an **Auto-Fail Gate** for baseline configurations, executing a branching route transition:

- **Branch A (Correct Text Input Matrix):** Unlocks "Friendly Markup Mode" inside the Committee Mini-Dungeon. The Extremist functions as a temporary modifier asset, leading to a smooth progression return loop with their entire faction's block locked in.
- **Branch B (Incorrect/Timed-Out Input Matrix):** Sets the global variable to "Hostile Markup Mode." The Committee Room instantiates as a high-aggression gauntlet, forcing micro-negotiations with hidden backbenchers. The Extremist serves as the final floor boss block.

3. THE POKÉMON-STYLE CAPITOL MAP ARCHITECTURE

The map below demonstrates how the classical, grid-based topographic layouts of 8-bit/16-bit handheld environments translate directly into the structural floorplans of Washington bureaucracy. This layout represents **The House Wing & Central Rotunda Hub Complex**.



3.1 Map Map Legend & Tile System Rules

- | | |
|--|---|
| [P1] Player Spawn / Current Orientation Anchor | [LEADER X] Coalition Head Intermediation Nodes |
| (M) / (B) / (R) Static, Non-Moveable NPC Sprites | [X] / # Asset Hitbox Gates / Open Navigation Thresholds |
| [C1] / [C2] Committee Room Instance Sub-Vectors | ==== Collidable Structural Wall Geometries |

4. NPC DISTRIBUTION & MOVEMENT ARRAYS

4.1 Population Densities & Sprite Matrix Rules

To simulate a dense congressional atmosphere without rendering independent movement logic for hundreds of unique entities—which would degrade hardware ticks on lower-spec Chromebook configurations—the system utilizes a **"Cluster Anchor System."** All decorative constituent entities (*M*, *B*, *R*) are anchored permanently to their specific coordinates, blocking pathways and forming a literal visual maze around their respective Coalition Leaders.

- **The Floor Chamber:** Features exactly 45 static placeholder character sprites grouped into geometric rings around 3 interactable Leader units.
- **The Cloakrooms & Side-Offices:** Populate with hidden, roaming *Rogue Nodes* that utilize basic random 2-tile ping-pong wandering routines to simulate backroom wheeling and dealing.

4.2 Movement Arrays & Collision Triggers

The layout uses classical rigid cell-based bounding boxes (*16 times 16* pixel tiles). The player sprite navigates linearly across open paths. Entering the coordinate buffer zones directly surrounding any Coalition Leader triggers immediate movement restriction, forcing UI dialog activation loops.

Architectural Gate Trigger Script

If `Player_Coordinates` intersect `Extremist_Ambush_Zone` AND `175 ≤ Vote_Counter < 218`, the system freezes user keyboard map inputs, instantiates the Extremist Sprite overlay directly in the player's path vector, and forces an immediate combat interface switch. Escape keys or pathing workarounds are strictly disabled at compilation levels.

5. THE CURRENCY LOOP & STATE SAVING

The Rotunda serves as the dedicated spatial anchor linking the House and Senate chambers. It contains the central **Influence Store Vendor Node**, allowing players to trade acquired Influence integers for execution modifiers (e.g., **Hearing Aids**, **Talking Memos**, or **Notebook Clues**). It also houses the primary `Save_Station` terminal, which exports a lightweight JSON state string containing flags for passed checks, gathered notebook entries, and currency scores directly to local storage.